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Haoming Yang Ph.D.

Research Interest:

There are mainly two sides to my research.

- Foundational Deep Learning: I apply the rich mathematical structure of **Stochastic and Partial Differential Equations** to model the dynamic processes inherent in neural network training, study the stochastic nature of optimization methods, and integrate physical principles directly into the learning process. These equations can be applied in deep learning to provide valuable insights to develop **robust learning, continual learning, and sequence modeling** architectures.
- Deep Learning in Neuroscience: I study insect neuroscience and neuromotor controls from a deep learning perspective. Our goal is to understand how insects, with limited resources, perceive a complex changing environment and make rapid decisions. On this front, I develop **(semi)-interpretable deep learning algorithms** and collaborate with neuroscientists to understand how complex environmental information is gathered and processed by insects to direct their highly efficient and agile movements.

Education

Duke University — Durham, NC
Ph.D. in Electrical and Computer Engineering — 05/2026 *(expected)*
- Advisor: Prof. Vahid Tarokh
- Research area: Foundations of Deep Learning; ML in neuroscience

Duke University — Durham, NC
Master of Science in Statistical Science — 12/2022
- Advisor: Prof. David Dunson
- Research area: Interpretable Machine Learning in neuroscience

University of Illinois at Urbana Champaign — Champaign, IL
Bachelor of Science in Statistics (Highest Distinction)
Bachelor of Science in Engineering Physics (Highest Honor) — 05/2021
- Minor: Mathematics

Publications

1. **Haoming Yang***, Pramod KC*, Panyu Chen, Hong Lei, Simon Sponberg, Vahid Tarokh, and Jeffrey Riffell. Neuron synchronization analyzed through spatial-temporal attention. ***Frontiers In Computational Neuroscience***, 2025.
2. Ali Hasan*, **Haoming Yang***, Yuting Ng, and Vahid Tarokh. Elliptic Loss Regularization. In: The Thirteenth International Conference on Learning Representation ***ICLR***, 2025.
3. **Haoming Yang**, Ali Hasan, and Vahid Tarokh. Parabolic Continual Learning. In: The 28th International Conference on Artificial Intelligence and Statistics. PMLR, ***AISTATS*** 2025.
4. **Haoming Yang***, Ali Hasan*, Yuting Ng, and Vahid Tarokh. Neural McKean-Vlasov Processes: Distributional Dependence in Diffusion Processes. (**Oral**) In: The 27th International Conference on Artificial Intelligence and Statistics. PMLR, ***AISTATS*** 2024.

Preprints

1. Zihao Wu, Juncheng Dong*, **Haoming Yang***, Vahid Tarokh. S2TX: Cross-Attention Multi-Scale State-Space Transformer for Time Series Forecasting. *Under Review*.
2. Barredo, Elina*, **Haoming Yang***, et al. Body size and light environment modulate flight speed and saccadic behavior in free flying *Drosophila melanogaster*. *Under Review*.
3. **Haoming Yang***, Marko Angjelichinoski*, Suya Wu, Joy Putney, Simon Sponberg, and Vahid Tarokh. Cross-subject Mapping of Neural Activity with Restricted Boltzmann Machines. *Under Review*.
4. **Haoming Yang**, Steven Winter, Zhengwu Zhang, and David Dunson. Interpretable AI for relating brain structural and functional connectomes.

In Progress

Guiding LLM Reasoning with Logic Flows
Developing novel methods to guide LLM to reason with logic by guiding LLM with canonical logic flows.

Incontext Learning of Functionals
Researching transformer's in-context learning ability on distribution flows. Developing mathematical model and architecture to study in-context generative models.

Interpretable Models in Neuroscience
Architect deep learning model to model neural representation that leads to linear behavior in agile insects.

Interpretable Models for Neuromotor Control
Formulate neural network system to uncover neural to motor connections. Identify motor mechanisms for spontaneous, agile movement under odor and visual stimuli.

Selected Experience

Meta | Data Science Intern — 2025.05 - 2025.06

- Developed a novel regression-based, robust, automated, data quality check. Designed a causal inference-inspired volatility metric to flag potential error data. These quality filters greatly improve the robustness of downstream SoT data application.

Morgan Stanley | Joint Researcher — Current

- Collaborating with scientists at Morgan Stanley researching application of SDE/PDE in various areas of machine learning. Published on top machine learning conferences.

IRisk Lab, NLP Risk Management Group | Researcher — 2020.09 - 2021.07

- Developed Natural Language Processing Tools for Actuarial Science Applications including an unsupervised method to identify businesses' related industries.

Tencent | Data Science Intern — 2020.06 - 2020.09

- Led the design and implementation of internal data analysis tool to capture product demographic similarity. Conducted multiple A/B tests and market analysis on behavioral data. Improved user stickiness by 30%.

Awards and Honors

Oral Presentation (Top 5%) in AISTATS 2024 — 2024.01
Duke Fellowship — 2023.01
Undergraduate Highest Honor & Highest Distinction — 2021.05

Professional Services

Reviewer for Conference: *ICLR 2025, 2026, AAAI 2026*
Reviewer for Journal: NueroImage
Teaching Assistant, *ECE 685D: Introduction to Deep Learning* — Duke University
(2022, 2024, 2025)

Skills

Languages: Chinese (Fluent), English (Fluent), French (Intermediate)
Programming Languages: Python, R, SQL.
Developer Tools: PyTorch, Jax, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn

Mentoring

Zihao Wu: Previous master student at Duke, now Ph.D student at Duke.
Junyi Liao: Ph.D student at Duke.
Matthew LaRosa: Master student at Duke.
Panyu Chen: Master student at Duke.